

Federico Iván Bilotti Gameplay Programmer

 LinkedIn  CABA, Argentina  GitHub

PROFESIONAL PROFILE

Virtual Simulation Programmer with strong experience in Unity and C#, and basic knowledge of Python and C++. Motivated and adaptable, both in team environments and when working independently. I have participated in various projects with the goal of gaining experience and meeting set objectives.

I also stay up to date with the latest technologies, continuously learning and exploring through documentation, online communities, and educational content. Some of the topics I've recently explored include Unity DOTS, Clean Architecture, and Docker.

PROJECTS

09/2023 – 12/2024	Denos - Unity Programmer , Final Project - Unity HDRP  First-person body horror game in which you subject yourself to unimaginable tortures by infecting yourself with alien larvae, causing mutations in your body that you can use to solve puzzles in an organic world controlled by an omnipresent entity. • Programmed for consoles (Xbox and PlayStation) and PC. • Created custom editor tools in Unity (EditorWindows and CustomEditors) to simplify and accelerate the configuration and design of mini-games. • Developed inverse kinematics (IK) animations for body mutations and world interactions using Unity's Animation Rigging package. • Built a scalable save system with support for additive scenes. • Extensive use of Unity Profiler and Memory Profiler to diagnose bottlenecks; reduced texture memory usage by 75% and draw calls by 90%. • Designed and implemented a Dependency Injection system, ensuring stable runtime performance with no FPS drops. • Implemented 10+ design patterns (Singleton, Service Locator, Observer, Factory, Strategy, State, Builder, Adapter, Dependency Injection, among others) to maintain flexible, reusable, and testable code.
02/2025 – 02/2025	Parallel Systems Programmer – Pathfinding-Astar , Unity C#  Implementation of the AStar pathfinding algorithm and a Custom Native Container (MinHeap), using Unity Job System and Burst Compiler. • Implemented pathfinding on a 100×100 grid in parallel using Unity's Job System and Burst Compiler, reducing execution time by approximately 99%. • Developed a custom native container (MinHeap), achieving an ~85% reduction in extraction operation cost. • Applied data- and object-oriented programming principles to ensure scalability and maintainability.
06/2025 – 06/2025	Backend developer - API Video Game Explorer , C# ASP.Net  Development of a RESTful API in ASP.NET  Core for video game management, applying best practices in design, testing, and maintenance. • Clean Architecture and Repository Pattern. • Dapper for lightweight, high-performance data access. • FluentValidation for request validation and centralized error handling. • JWT Authentication & Authorization (access & refresh tokens) with role support. • API Versioning. • Health Checks endpoints for service monitoring and readiness. • Swagger (OpenAPI) UI for API documentation and testing. • Docker & Docker Compose for development and integration testing. • Integration Testing with xUnit against ephemeral Docker databases.

EDUCATION

2021 – 2024 CABA, Argentina	Designer and Programmer of Virtual Simulators , Escuela Da Vinci  Unity, C#, C++, Object-Oriented Programming (OOP), Design Patterns, Artificial Intelligence, LINQ, Unreal Engine
2015 – 2019 CABA, Argentina	Bachelor of Visual Arts , Colegio y Oratorio San Francisco de Sales 

SKILLS

• Unity	• C#	• DOTWEEN	• DOTS
• SQL	• Inverse Kinematic	• Git / GitHub	• C++
• Python	• ASP.Net Core	• API Rest	• SOLID
• Docker	• Testing / xUnit	• Trello	

LANGUAGES

• Spanish Native	• English B1
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